Synopsis Report on Mini Project

FitAI: Intelligent Vision-Based Personal Fitness Coach

Submitted in partial fulfilment of the requirements of the degree of Bachelor of Engineering in Computer Science and Engineering

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Synopsis Approval

This synopsis entitled **"**FitAI: Intelligent Vision-Based Personal Fitness Coach**"** by student names is approved for the degree of Bachelor of Engineering in Computer Science and Engineering (AI & ML).

Examiners

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Supervisors

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HOD

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Principal

_____

Date:

Place

Declaration

We hereby declare that the written submission presented in this synopsis report represents our original ideas, research, and developmental efforts. The core concepts, system architecture, and implementations detailed within this document accurately reflect the scope of our own work.

Wherever the ideas, words, frameworks, or code snippets of others have been utilized—including the integration of third-party APIs, machine learning models, or external software libraries—they have been adequately cited and comprehensively referenced in accordance with standard academic practices. We have ensured that proper permissions were obtained where necessary and that all open-source guidelines have been respected.

Furthermore, we declare that we have strictly adhered to all principles of academic honesty and integrity throughout the development of this project. We have not misrepresented, fabricated, or falsified any data, testing outcomes, facts, or source materials in this submission. The capabilities and performance of the system described herein are a truthful and accurate representation.

We fully understand that any violation of these ethical standards—including plagiarism, unauthorized use of intellectual property, or data falsification—constitutes a serious breach of academic conduct. We acknowledge that such actions will give cause for severe disciplinary action by the Institute, and may also evoke penal action from external sources whose work has not been properly attributed or permitted.

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Abstract

FitAI is a sophisticated web-based application developed to democratize access to elite personal fitness coaching through artificial intelligence. In today's fitness landscape, working out without professional guidance often leads to improper form, resulting in severe joint and muscle injuries over time. Elite personal training, however, is expensive and inaccessible to the average person. This project focuses on designing an intelligent system that solves this problem using real-time computer vision and conversational AI.

The system utilizes TensorFlow.js and the MoveNet model via the device's camera to analyze skeletal joint angles, provide live form correction, and automatically count repetitions. Concurrently, it integrates Google Gemini's advanced LLM to act as a conversational coach, offering context-aware advice based on the user's goals, logged meals, and fatigue levels. The project is implemented using modern web technologies, ensuring that an adaptive, injury-aware, and highly personalized fitness experience is accessible to anyone with a browser. Overall, FitAI demonstrates the practical application of machine learning in real-world health scenarios, promoting safer and more efficient physical training.

Acknowledgements

We would like to express our sincere gratitude to our project guide, Devyani Gawade, for their valuable guidance, continuous support, and encouragement throughout the development of our mini project titled "FitAI: Intelligent Vision-Based Personal Fitness Coach." Their technical insights and suggestions have greatly contributed to the successful completion of this work.

We are also thankful to the Head of the Department and all the faculty members of the Department of Computer Science and Engineering, [Your Institute Name], for providing us with the necessary facilities and a supportive academic environment. We would like to acknowledge the cooperation and support of our peers during the course of this project. Finally, we express our sincere thanks to our parents for their constant encouragement and support.

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Chapter 1 Introduction

In the modern digital era, health and fitness have become paramount, yet the barrier to safe, professional guidance remains high. Individuals often rely on static workout plans or generic fitness applications that cannot adapt to their specific biomechanics, current fatigue levels, or age-related limitations. The project "FitAI: Intelligent Vision-Based Personal Fitness Coach" is developed to bridge this gap. It provides users with an intelligent, browser-based coach that actually "sees" their movements. The system is designed to be highly interactive, processing live camera feeds to ensure safety and precision during workouts, while handling daily accountability tracking like hydration and progressive overload

1.1 Importance of project

The importance of this project lies in its ability to prevent physical injury while democratizing health. By providing live skeletal angle-tracking and an injury-aware exercise filtering database, it drastically reduces the likelihood of gym-related injuries. It ensures fitness is accessible and tailored for all demographics, including providing low-impact modifications for senior citizens.

1.2 Motivation

Existing fitness systems are predominantly text-based or static video libraries. They lack the ability to correct a user who is performing a squat or push-up incorrectly. The motivation behind FitAI is to bring the expertise of a human personal trainer into an accessible web application. The aim is to create a fast, reliable platform that actively monitors form, dynamically autoregulates future sets based on exertion feedback, and motivates the user seamlessly.

1.3 Organization of the dissertation report

- **Chapter 1:** Introduction - Overview of the project
- **Chapter 2:** Literature Survey - Study of existing fitness and AI vision tracking
- **Chapter 3:** Design Methodology - System design and working architecture
- **Chapter 4:** Tools to be Used - Technologies and frameworks
- **Chapter 5:** Conclusion and Future Scope - Summary and potential improvements

Chapter 2 Literature Survey

2.1 Overview of AI in Fitness

The integration of Artificial Intelligence in health and fitness has primarily revolved around predictive analytics, diet tracking, and step counting. However, the advancement of browser-based computer vision models has opened new avenues for real-time biomechanical analysis. Systems can now map human poses, tracking key points like shoulders, elbows, and knees, to analyze movement patterns without the need for expensive wearable sensors.

2.1.1 Need for Real-Time Processing

In physical training, delayed feedback is ineffective and potentially dangerous. If a user's spine is misaligned during a heavy lift, they need instantaneous correction. Therefore, the application requires lightweight, optimized models that can run directly on the client's device (edge computing) to eliminate server latency and protect user privacy.

2.2 Applications of Vision and LLMs

Computer vision models (like MoveNet or PoseNet) are used in sports science for performance analysis. Simultaneously, Large Language Models (LLMs) are used in digital health to provide personalized, human-like coaching interactions. FitAI merges these two applications, utilizing vision for physical correction and LLMs for psychological motivation, programming, and nutritional advice.

2.3 Limitations of Existing Systems

Current market solutions suffer from several limitations:

1. **Cost:** Apps offering true AI form correction often require expensive subscriptions or proprietary camera hardware.
2. **Inflexibility:** Standard apps do not dynamically adjust to daily user fatigue (autoregulation).
3. **Accessibility:** Most tools ignore demographics like the elderly or those with pre-existing injuries, offering generic routines that can exacerbate pain. FitAI overcomes this by incorporating dynamic difficulty adjustments and injury-specific filtering.

Chapter 3 Design Methodology

3.1 System Design

System Design and Architecture The FitAI system is engineered as a client-heavy, progressive web application to ensure real-time performance, cross-device compatibility, and strict user data privacy. To achieve a seamless integration of computer vision and conversational AI, the system architecture is divided into four highly synchronized operational modules:

1. Client-Side Presentation Layer (User Interface) Developed using React.js, this layer is responsible for managing the dynamic application state and rendering the user interface. It handles seamless navigation between the user dashboard, the active workout camera view, the AI chat interface, and daily habit trackers (such as the hydration logger).

2. Vision-Based Inference Engine (Biomechanical Tracking) This is the core technical component of the system. Instead of sending video data to a server (which introduces latency and privacy risks), the system utilizes TensorFlow.js to run the MoveNet pose estimation model directly in the user's browser via edge computing.

- **Frame Processing:** It captures the live camera feed and processes frames at high speeds to detect 17 key skeletal points (shoulders, elbows, hips, knees, etc.).
- **Angle Computation:** The engine uses trigonometric algorithms to calculate the angles between these keypoints in real-time. By comparing these live angles against standard biomechanical thresholds, the system can accurately auto-count repetitions and instantly flag improper form (e.g., warning a user if their back is curving during a deadlift).

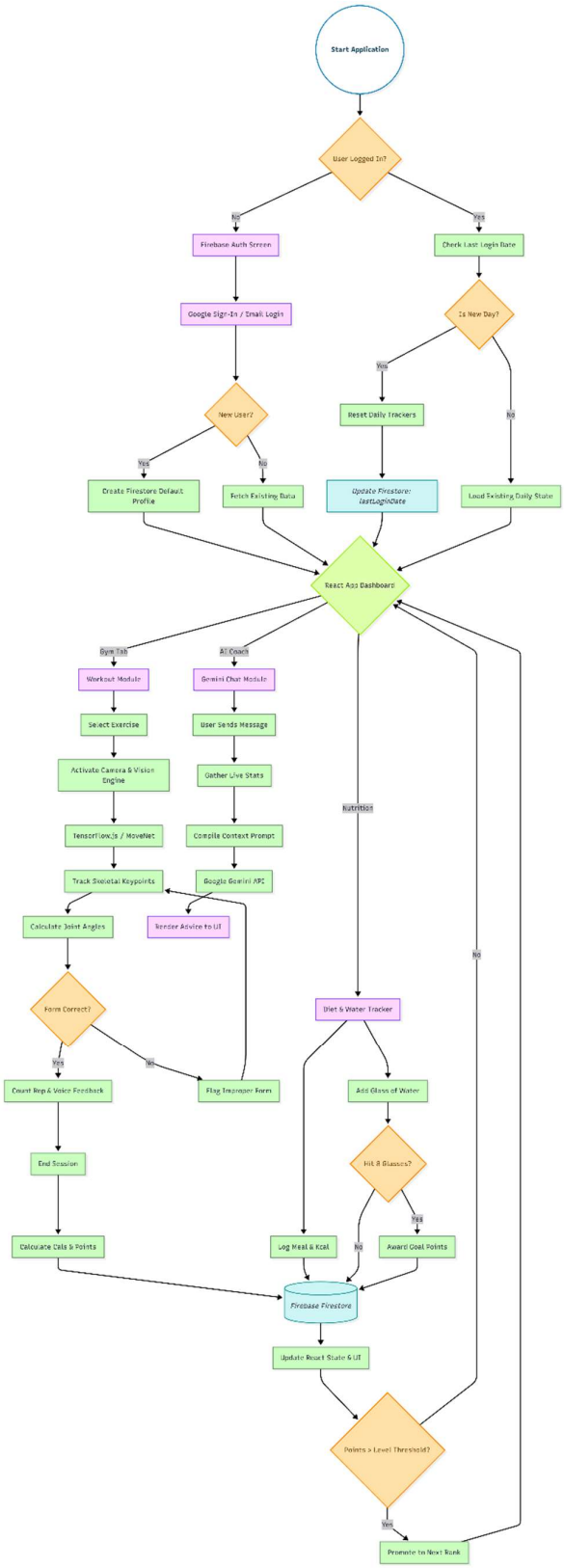
3. Cognitive AI Engine (Conversational Coach) This module acts as the "brain" of the application, powered by the Google Gemini API. It takes structured data from the user—such as their age, logged injuries, daily dietary input, and immediate fatigue levels—and processes it using complex prompt engineering. The LLM then generates highly contextual, conversational advice. It is also responsible for "Autoregulation," meaning it mathematically adjusts the required sets, reps, and weights for future workouts based on the user's real-time feedback.

4. Cloud Backend & Data Persistence Layer To maintain user progress across multiple sessions and devices, the system connects to Google Firebase.

- **Authentication:** Manages secure user login and session tokens.
- **Firestore Database:** A NoSQL database structure that stores the user's static profile data (height, weight, goals), dynamic daily logs (water glasses drank, meals eaten), and gamification metrics (workout streaks and fitness scores).

System Workflow The operational workflow begins with secure user authentication, followed by the retrieval of their historical data from Firebase to populate the dashboard. When a user initiates a workout, the Vision Engine activates the local camera, running continuous real-time inference to guide the physical session. Post-workout, the results are passed to the Cognitive AI Engine for analysis and feedback, and the final updated metrics are securely synced back to the Firebase cloud database.

Flowchart:


Start Application
User Logged In?
No
Yes
Firebase Auth Screen
Check Last Login Date
Google Sign-In / Email Login
Is New Day?
New User?
Yes
No
Reset Daily Trackers
Create Firestore Default Profile
Fetch Existing Data
Update Firestore: lastLoginDate
Load Existing Daily State
React App Dashboard
Gym Tab
AI Coach
Nutrition
Workout Module
Gemini Chat Module
Select Exercise
User Sends Message
Activate Camera & Vision Engine
Gather Live Stats
TensorFlow.js / MoveNet
Compile Context Prompt
Track Skeletal Keypoints
Google Gemini API
Calculate Joint Angles
Render Advice to UI
Form Correct?
Diet & Water Tracker
Count Rep & Voice Feedback
Flag Improper Form
Add Glass of Water
End Session
Hit 8 Glasses?
Calculate Cals & Points
Log Meal & Kcal
Award Goal Points
Firebase Firestore
Update React State & UI
Points > Level Threshold?
Promote to Next Rank

Chapter 3.2 Tools to be used

The development of the FitAI: Intelligent Vision-Based Personal is based on a structured technical approach that includes frontend, backend, database, and intelligent processing components. These tools work together to ensure efficient performance and a smooth user experience.

Frontend (User Interface)

The frontend is developed using **React.js** and **Vite** to provide a highly responsive, single-page application experience. CSS and modern design principles are used to create an intuitive dark-mode aesthetic.

Vision & Processing Layer

TensorFlow.js (MoveNet): Used for real-time skeletal tracking. It detects 17 key body points, allowing the system to calculate mathematical angles between joints to verify correct exercise posture.

AI & Cognitive Layer

Google Gemini API: Integrated to serve as the conversational brain of the application. It analyzes user input, daily logs, and workout history to provide intelligent, contextual feedback and training adjustments.

Database & Backend Services

Firebase: Utilized for secure User Authentication (Google Sign-In) and Firestore Database to manage user profiles, daily water tracking, workout history, and gamification streaks.

Deployment & Development Tools

- **Netlify / Vercel:** For continuous deployment and global hosting of the web application.
- **Visual Studio Code (VS Code):** Used as the primary integrated development environment.
- **GitHub:** For version control and repository management.

4.Results:

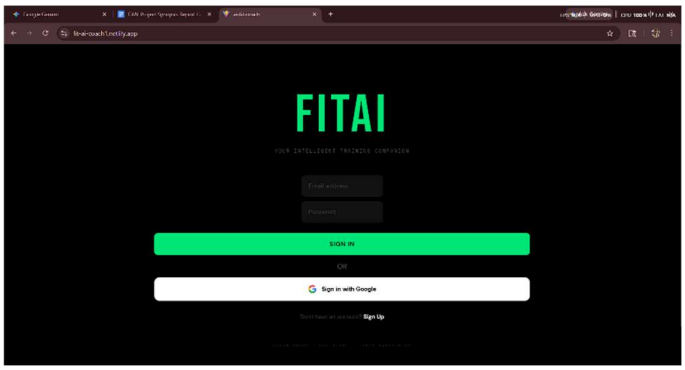


Fig1.Login Page

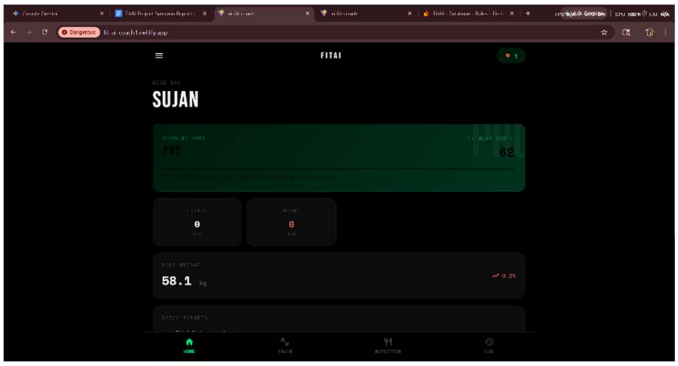


Fig2.Dashboard

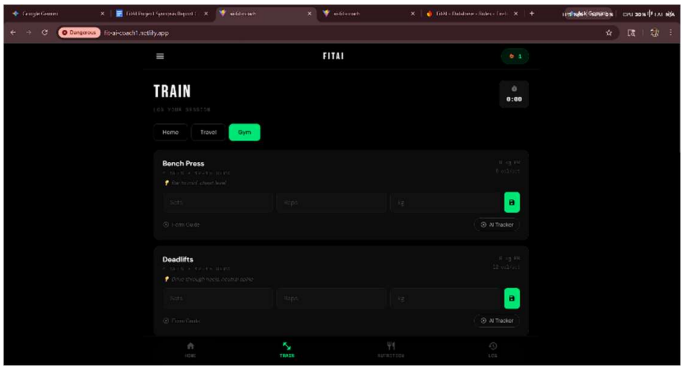


Fig3.Exercise

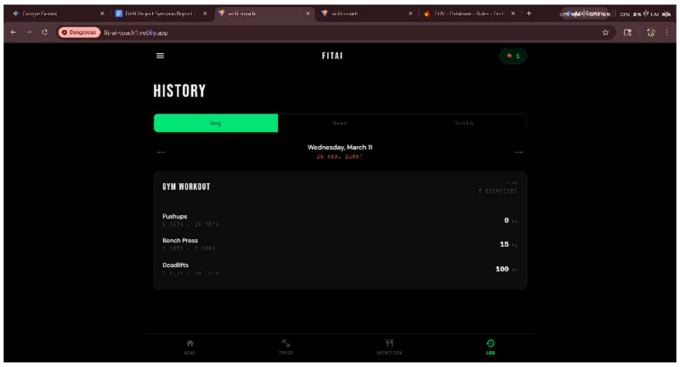


Fig4.History

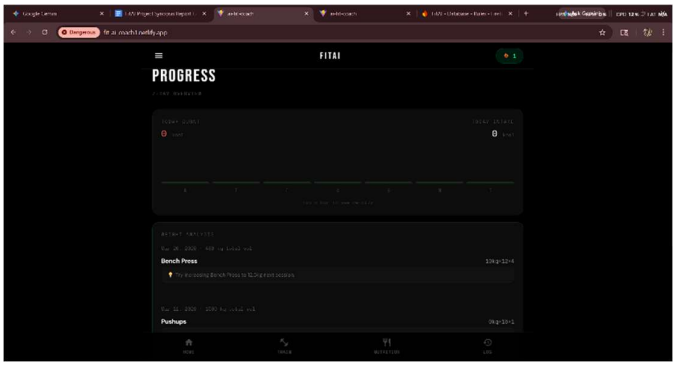


Fig5.Progress

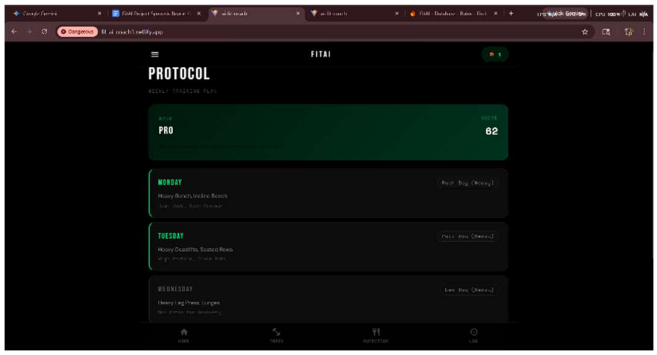


Fig6.Weekly planner

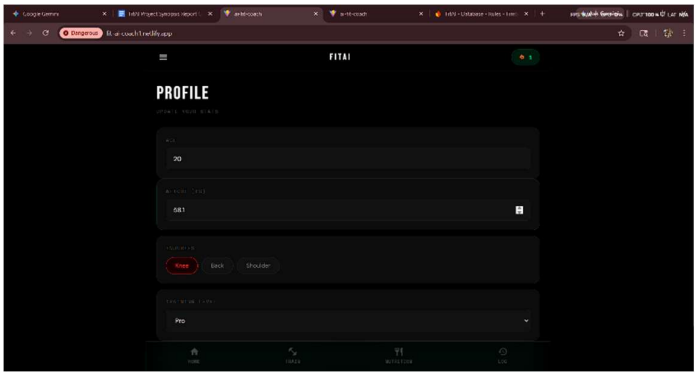


Fig 7.Profile

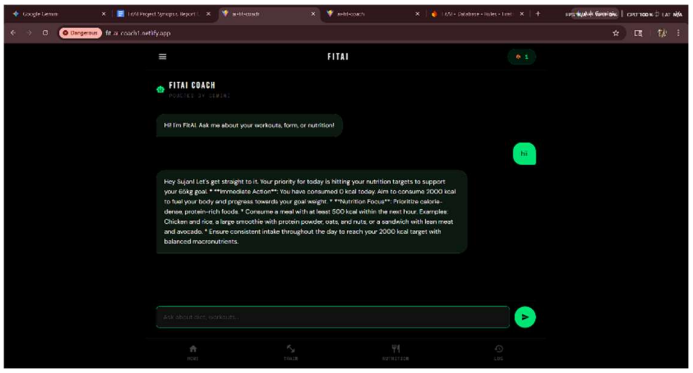


Fig 8.Ai Coach

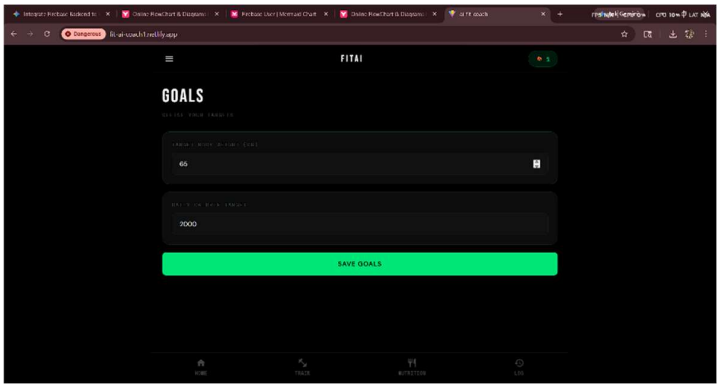


Fig 9.Set Goals

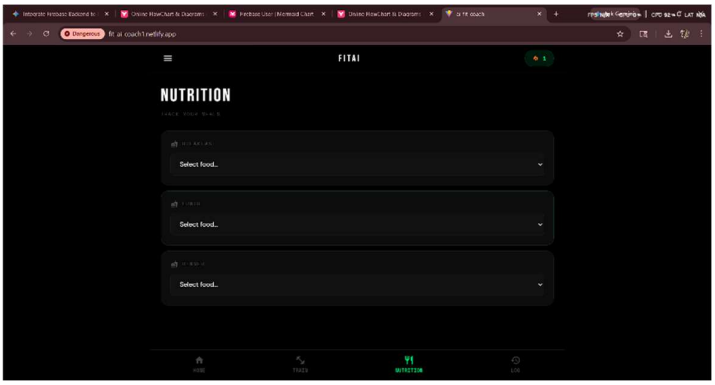


Fig 10.Kcal Logger

Chapter 5: Conclusion and Future Scope

Conclusion:

The project "FitAI: Intelligent Vision-Based Personal Fitness Coach" successfully demonstrates a highly capable, intelligent alternative to traditional personal training. By successfully integrating TensorFlow.js for live pose estimation and the Gemini API for conversational coaching, the system effectively corrects user form, tracks progress, and adapts to individual physical limitations. The utilization of web technologies ensures the platform remains highly accessible, requiring no specialized hardware. FitAI successfully achieves its objective of making safe, personalized fitness guidance available to everyone.

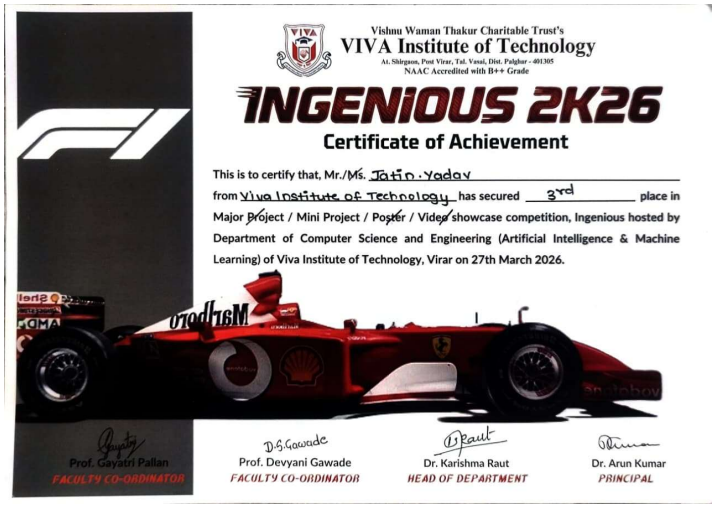
Future Scope:

The system can be further enhanced by incorporating a wider database of exercises and supporting multi-person tracking for group workouts. Future updates could include integration with smartwatches (Apple Health/Google Fit) to pull live heart-rate data, allowing the AI to adjust workout intensity based on cardiovascular strain. Additionally, the dietary tracking component could be upgraded with image-recognition, allowing users to simply take a photo of their meal to automatically log caloric and macronutrient data.

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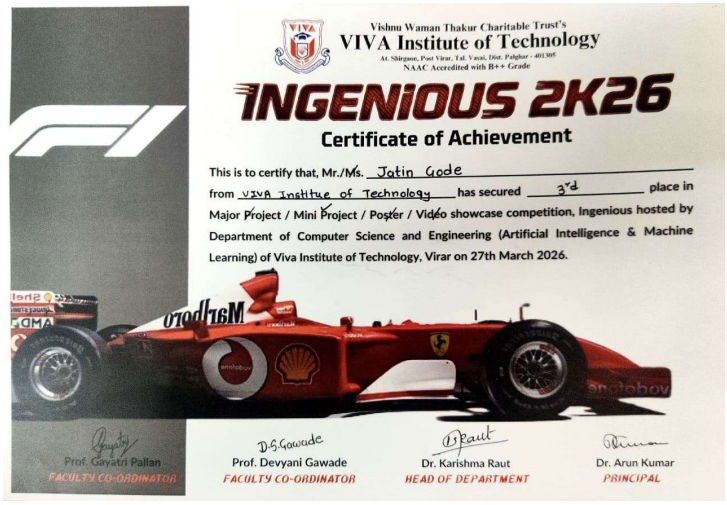
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Dr. Karishma Raut — HEAD OF DEPARTMENT
Dr. Arun Kumar — PRINCIPAL



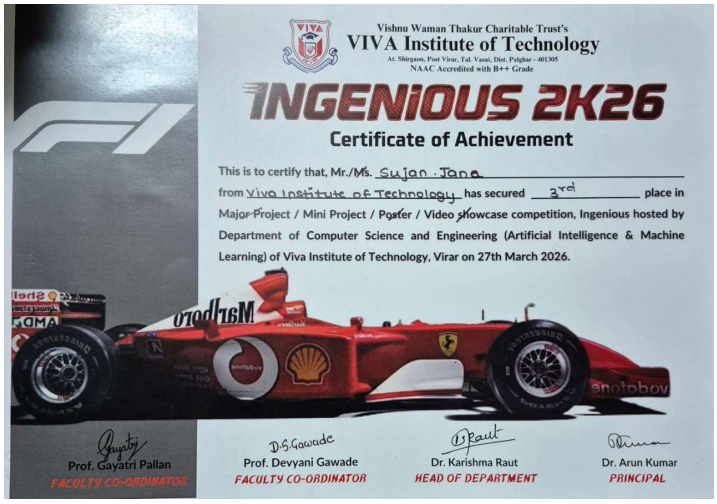
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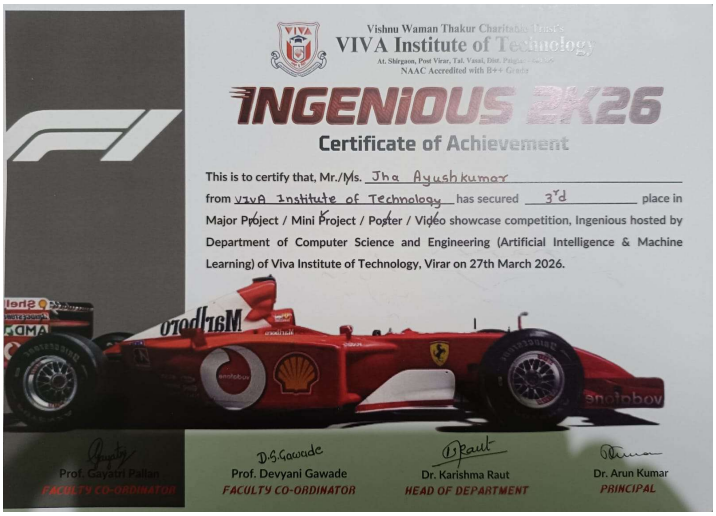
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